Blinded-V5: the LGI-based compute module

design rationale, construction, parameters, and measurements

local-mixing project

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Where it fits in the pipeline

The GSS gadgetization of a secret circuit C on n wires is a **5-stage** sandwich, each stage on $2n$ wires (data $0..n$, band/junk $n..2n$): (1) **slice** — a junk-guard zero-slice keyed on the band; (2) **seed the band wires** — each band wire set to $x_i \wedge \neg x_j$ from the honest inputs; (3) **compute** — realise A (the sliced sandwich of C) on the $2n$ wires so the data half carries A's output and the band is only read; (4) **re-seed / re-randomise the band**, interleaved with the compute; (5) **re-slice** — the closing junk-guard slice.

Blinded-V5 is a drop-in alternative for stages 3–4 (the compute and its band re-randomisation), replacing the drip `route_fire` compute. It takes A on n wires and emits an equivalent A' on $2n$ wires whose body is a cloud of long masking identities with A's gates threaded through by a masked read. Stages 1, 2 and 5 are unchanged. Source: `src/preprocessing/blinded_v5.rs`; the Markdown twin of this document is `BLINDED_V5_LGI_DESIGN.md`.

1 Why this design — the rationale

The compute stage must move the intermediate state as far as possible from the plaintext computation A while keeping A' exactly equivalent to A on the data wires. The design is a chain of forced moves.

(a) Long identities are the only thing that moves the state. Across the HD and affine heatmaps, short local edits barely displace the state; the ridge of recoverable structure only recedes when the masking identity woven through a wire is *long* — comparable to the circuit itself.

(b) The only long identity that moves the state is the commutative structure of an identity with a single active wire. A run of gates all targeting one active wire w and reading only controls is an identity iff the increments XOR to zero, and *all its gates commute* — which lets it be arbitrarily long and freely interleaved.

(c) Entangle identities on different active wires by sharing the control wires — the band. Every identity targets a data wire and reads only the band ($n..2n$), so identities on different active wires also commute; a band wire participates in the masks of many data wires at once.

(d) Add random updates of the control wires from the active wires. Band updates (half live-data-controlled $b \oplus= \mathit{data} \wedge \mathit{aux}$, half band-only) **re-randomise** the controls and **break naive commutation-back**, without ruining correctness (each identity is closed before the update or re-derived across it).

(e) Embed A inside one circuit-spanning identity via hidden unmasking. Each gate fires from inside the mask: its operands are momentarily and reversibly unmasked into a linear combination of band wires (never bare), it fires as an expansion over those masked wires, then re-masks.

(f) Co-sample masking, band updates, and A-placement together. Building them in one pass — rather than a fixed scaffold threaded afterwards — lets every A-gate be straddled by a real, rerand-protected LGI opened exactly when the gate needs it (§3), which makes the firing-hiding complete at no cost.

2 The construction

A has n wires and m gates; A' has $2n$. Let u_w be the uses of wire w in A and w_w its writes.

Masking atom. $\mathtt{g57}(w, x, y) = w \oplus= 1 \oplus (\neg x \wedge y)$ (data target, band controls). A disjoint-pair LGI $\bigoplus_i \mathtt{g57}(w, cy_{2i}, cy_{2i+1})$ is a deg-2 mask; a `g57` and its reverse **linearise**, $\mathtt{g57}(w, r_1, r_2) \oplus \mathtt{g57}(w, r_2, r_1) = w \oplus r_1 \oplus r_2$ (AND monomials do not). Disjoint pairs, not a K-cycle (which would telescope to a bare operand); $K \geq 2$.

Co-sampled forward pass. The LGIs, the band updates and A's gates are emitted together. Each active wire w gets $u_w + 1$ LGIs, $\leq$ `max_open` open at once — the SAME masking budget as a fixed scaffold, so the same statistics, at no cost. Of w's opens, w_w are **STRADDLE** opens generated on demand as A's gates on w are placed, and the rest $(\,\mathrm{reads}_w + 1\,)$ are **FILLER** opens. A's gates are placed in a data-hazard-valid order (a read after every earlier write of a wire, a write after every earlier read; same-target XOR writes commute, no WAW edge).

At each A-gate on active wire c: the **masked read** linearises each control (emit the reverse `g57` of every open mask so the wire carries $\mathit{operand} \oplus \rho$, ρ a linear XOR of band wires; a fresh-pair top-up keeps ρ non-empty $\Rightarrow$ no bare read), realises $c \oplus= \mathit{comp} \oplus \mathrm{lit}(a) \wedge \mathrm{lit}(b)$ as $(a' \oplus \rho_a)(b' \oplus \rho_b)$ over only the masked controls and band wires (0/1/2 controls, all polarities), then de-linearises; and the **hidden firing** of §3 straddles one of c's LGI opens.

Band re-randomisation is woven through the same pass: `rerand_level` STRADDLE updates (close the masks reading the updated band wire first — thins masking past a ≈ 1024 knee) and `rerand_repair` REPAIR updates (re-derive each such mask across the update so it stays open — no thinning); being in the same pass, it protects the straddle opens automatically.

Correctness is verified exhaustively over all 2^n inputs $\times$ many band settings for $n \leq 6$, $K \in 2..5$, all `max_open` and both rerand kinds (`scratchpad/v6`: 891 gadgets, 0 mismatches, plus a 360-case reorder-preserves-A check), and end-to-end in `gen_sandwich_gadget` (forward + reverse-honesty sample-verify PASS).

3 Hidden firing

An *atomic*, mask-restoring gate module would leak which gate fired: the active wire's before/after XOR across it equals the true increment $\Delta = comp \oplus \mathrm{lit}(a) \wedge \mathrm{lit}(b)$. That per-module increment is a robust invariant that survives the downstream mixing, so an adversary who segments the mixed circuit at module boundaries reads off A's gate list.

The fix uses the abundant masking gates on the *same* active wire: split the fire and emit one of c's LGI **opens between the halves**. It toggles c mid-fire and stays open past the module, so the net XOR on c is $\Delta \oplus$ (secret band-mask) — never the bare increment. Because everything is co-sampled, the straddling LGI is a real, rerand-protected scaffold LGI opened exactly when the gate is placed, so **every** gate is covered, reusing the wire's $u_w + 1$ budget at **no size cost** (measured: 7920/7920 gates on the n=128 sandwich). Earlier attempts — A-gate weaving (∼23%), an injected firing mask (100% but +43% size), straddling fixed-scaffold slots (60–70%, tail slot-starved) — are superseded by generating the slot on demand.

4 Parameters and why

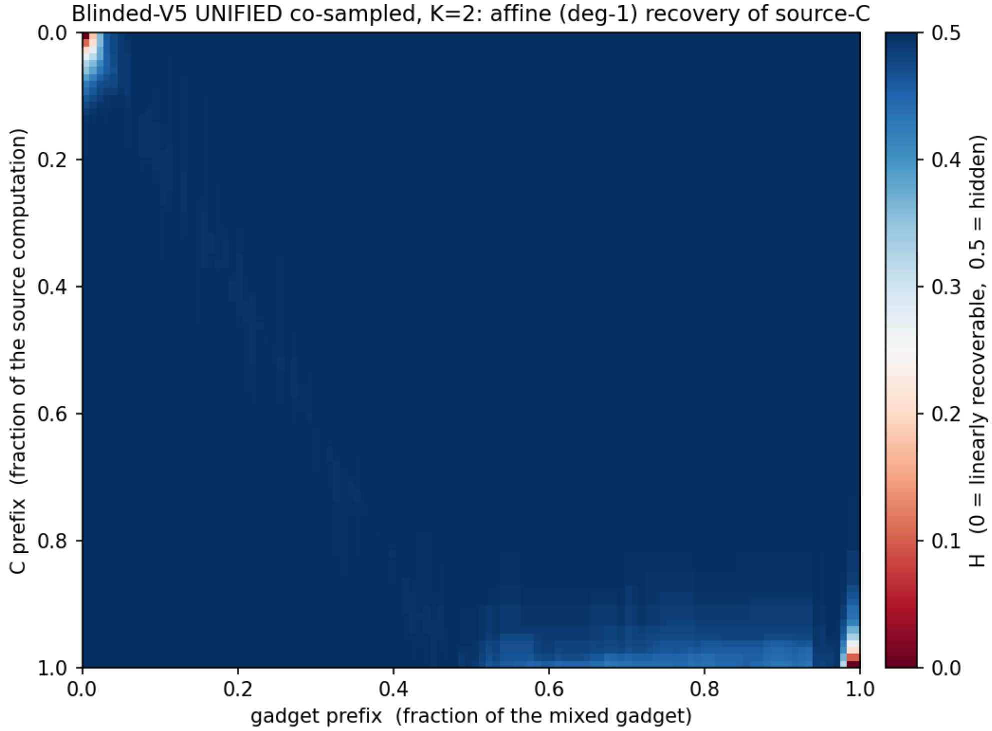
knob	prod.	meaning	why this value
K	**2**	band wires per LGI = per-LGI **mask width** ($\lfloor K/2 \rfloor$ pairs; $\lvert\rho\rvert \approx \texttt{max\_open} \cdot K$) — not the temporal length	affine/deg-2-neutral across K (§5); smallest is best. Size ∼linear, read cost quadratic in $\lvert\rho\rvert$, so large K explodes. Odd K wastes a wire.
`max_open`	**3**	rolling cap on open LGIs per wire; read-mask width floor	wider ρ = more local hiding, read cost quadratic; 3 is the knee.
`rerand_level`	**1000**	STRADDLE band updates	at the safe side of the ≈ 1024 thinning knee.
`rerand_repair`	**3000**	REPAIR band updates (no thinning)	band turnover on top of straddle at no masking cost.

K sets each LGI's mask *width*, not its temporal length — the identity's circuit-spanning length is a property of the whole scaffold (§1e).

5 Measurements

On the current co-sampled build (n=128, $|C| = |D| = 3000$, K=2); they match the pre-firing-fix baseline (the hidden-firing change did not move them).

5.1 Affine ridge

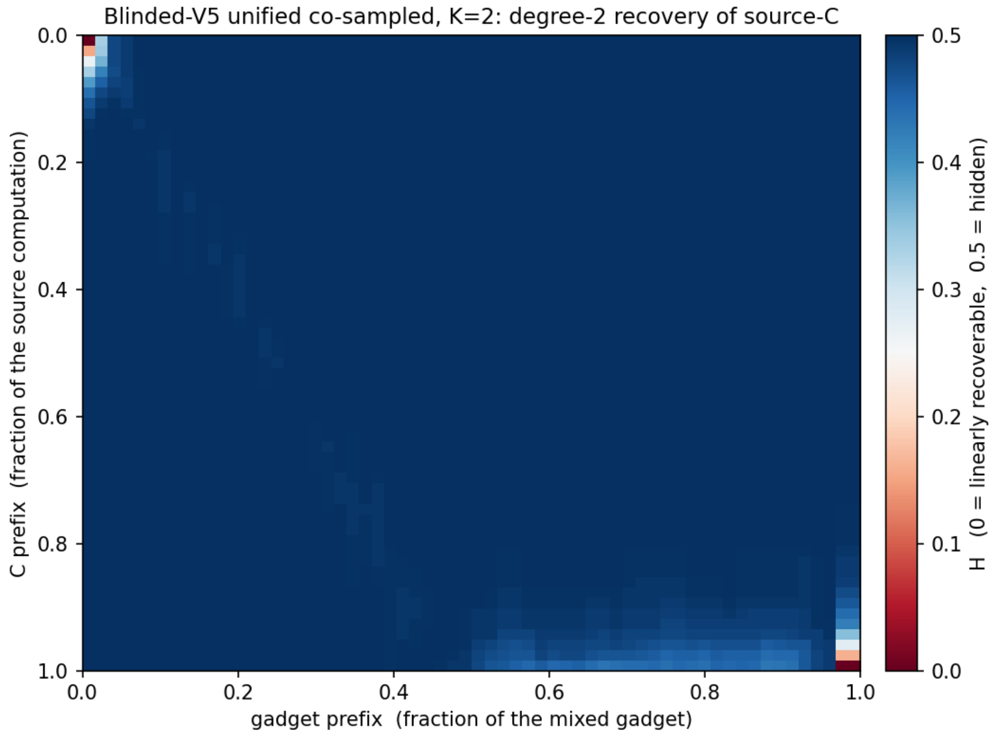


meanH ≈ 0.498, ridge depth ≈ 0.05; the interior ($\sim$78% of rows) is fully hidden ($H \approx 0.49$), and the only recovery is a thin fringe at the two endpoints.

5.2 The exposed fringe is public I/O — input segments are not leaked

The $\approx$6–10% of source-C segments that are linearly exposed (min-$H < 0.25$–0.35) are the two corners of the diagonal. The **input side is masked as soon as the compute begins** — recovery is confined to the extreme top-left corner, i.e. the raw input port where x is public *before* any masking. The output side is the necessarily-delivered $C(x)$. So the input wire segments show a **lack of linear exposure** beyond the trivial public input; the boundary-cushion knobs measured no effect because the fringe is unavoidable public I/O, not a leak.

5.3 Degree-2 recovery



meanH ≈ 0.497, depth ≈ 0.05. The quadratic adversary reaches only ∼0.01 deeper than affine, into the same low-degree I/O fringe; the interior stays fully hidden. Degree-2 is flat across K, like affine.

5.4 Rerand: repair vs straddle ($n_A = 256$, affine deg-1)

straddle	repair	gadget	meanH	affDepth
0	0	670k	0.4979	0.0550
1000	**3000**	668k	0.4959	**0.0514**
8000	0	607k	**0.4878**	**0.0681**
0	8000	780k	0.4985	0.0549

Heavy STRADDLE thins (meanH ↓, ridge ↑, gadget shrinks); heavy REPAIR does not. Production $1000 + 3000$ sits at the baseline floor.

5.5 Through the GSS pipeline ($K = 2$ vs $K = 4$)

Pending — from the co-sampled `gss_mix.sh --gadgetization-mode blinded-v5` *runs (n=128, half-size $|C| = |D| = 3000$; grow-to-2× then hold ≈ 10 effs, 2-eff snapshot; then split, crossing, fcompress), $K = 2$ and $K = 4$: affine ridge and exposed-C count at the 2-eff snapshot and at the end, with stage sizes.*

6 Tradeoffs and the current choices

- **Mask width K vs size.** Larger K widens each mask, not the temporal length; hiding is flat in K while size grows $\sim$linearly and the read cost quadratically, so **K=2**; K=4 only for a wider read-mask.
- **Band turnover vs size (straddle vs repair).** STRADDLE thins past ≈ 1024, REPAIR grows the gadget; **1000 straddle + 3000 repair** balances at the baseline floor.
- **Firing-hiding is free** in the co-sampled build (it reuses the existing LGI budget); earlier fixed-scaffold approaches paid coverage (23–70%) or +43% size.
- **The residual fringe is I/O, not a knob** — no parameter removes the public input/output boundary.